



Domain 4: Accelerate Workload Migration and Modernization



Overview

The fourth and last exam domain of the AWS Certified Solutions Architect Professional SAP-C02 certification exam is all about migration in AWS. As a Solutions Architect, the onus is on you to properly migrate your on-premises applications and services onto the AWS Cloud. This is a daunting task as there are many intricacies involved; not just with technologies, but also with infrastructure design, personnel, and budget. Some companies would often perform a lift and shift migration (also known as rehosting) onto AWS while other organizations re-platform or re-architect their existing systems. AWS provides a wealth of services, tools, and features that you can leverage for your migration workloads and to accelerate the whole process.

This domain has the least amount of questions in the SAP-C02 with 20% coverage so therefore, you have to limit the time you spent studying the topics here. The questions in the actual exam revolve around these topics:

- Select existing workloads and processes for potential migration.
- Determine the optimal migration approach for existing workloads
- Determine a new architecture for existing workloads.
- Determine opportunities for modernization and enhancements.

In this chapter, we will cover the related topics for migration in AWS that will likely show up in your Solutions Architect Professional exam.

Planning Out a Migration

When planning to do a migration, there are a lot of factors that you need to carefully assess. It is common for companies to skip the planning stage and go right into migration, which in the end just becomes a lift-and-shift process. Although lift-and-shift is a legitimate migration strategy, it is not always the best one to go with. You are missing out on a lot of potential improvements and cost savings to your operations which stems from not being able to fully take advantage of the cloud. To establish a proper migration plan, we will answer the basic questions of who, what, where, when, why, and how. These are some of the questions that you need to answer to properly designate tasks and objectives to your people.

Who — Who are the stakeholders involved? Who will oversee the migration (SME)? Who will decide what to migrate and what migration strategy to use? Who will be responsible for which part of the migration? Who will manage the new infrastructure after migration?

What — What is the business objective? What will you be migrating? What is the strategy for the migration? What tools will you be using for the migration? What is the timeline for the migration? What possible issues can come up before, during, and after migration?



Where — Where will you migrate your applications to? Will it be on a managed service or on a virtual machine? Will there be refactoring involved and where exactly? Where will you start?

When — When will the migration be performed? When is the migration expected to end? When can you test your new infrastructure for production workloads? When will the old infrastructure be expected to retire?

Why — Why will you migrate these applications? Why will you migrate these data? Why migrate this first before the others? Why aren't some applications being migrated? Why choose AWS over other platforms? Why migrate to EC2 or to RDS?

How — How will you perform the migration? How long will it take for you to migrate one application? One database? One complete service? How will you monitor your progress? How do you measure success after migration? How will you rollback after a failed migration? How will you manage the new infrastructure after migration? How much will this all cost you?

Migration Strategies

There are seven migration strategies, which are also known as the 7 R's, that we can perform in AWS. Each one has its own advantages and disadvantages.

Retire

This strategy is very easy to understand. If you don't need a certain application anymore then disregard it. Start by determining which components are essential to your business and which are not. This way, your migration process will only involve the important parts of your system, thereby reducing cost, effort, and time for everyone.

Relocate

Basically, this migration strategy is the act of relocating your proprietary virtualization platform from your on-premises network to the cloud. This is perfect if the existing license of your on-premises platform is supported in the cloud and you want to maintain a low to medium operational complexity in managing your infrastructure. You have to assess both the cost and the operational efficiency gains first via technology replatform against the required effort to re-platform or re-architect your stack while considering migration deadlines. For example: if your on-premises servers are running in a VMWare vSphere platform, you can consider relocating your virtualized servers to AWS using the VMWare Cloud on AWS service.

Rehost

The simplest method of migrating to AWS is to move your applications without changing them, essentially a "lift-and-shift" scenario. A common example for this is when you are moving your legacy web servers from on-prem onto EC2 instances. You treat the EC2 instances as if they were your own servers, thereby not



modifying any aspect of your application. This strategy is a quick and easy way to get things running in the cloud without much repercussion. Although you do not make use of all of AWS's advantages, you still receive some such as cheaper infrastructure pricing options, some elasticity and scalability for your VMs, as well as basic security and network services.

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Another not so common but possible application for re-host is when a customer wants to move a certain application to AWS but AWS currently doesn't support it as a native service. For example, you want to move a database to AWS but RDS does not support its engine or engine version, then you will have to use EC2 to run your database.

Replatform

With re-platform, you are utilizing new services to host your applications without changing the core of it. These services usually provide some form of support or feature that reduces management overhead. Examples would be AWS Elastic Beanstalk for hosting web applications and Amazon RDS for hosting your databases. The main benefit of re-platforming is to achieve increased savings and agility for your workloads.

i Solutions Architect Professional Exam Notes:

You might encounter a lot of PaaS scenarios in your exam, including services such as Elastic Beanstalk, RDS, ECS and many more. If the scenario requests that there should be less management overhead for the customer's application or database, look for services that give you that benefit. For example, with Elastic Beanstalk and ECS, they can quickly provision the resources you need with just a few clicks and they also support CI/CD deployment. They make it convenient for developers to apply their changes to production and quickly rollback if they encounter any issues. For RDS, these are the common maintenance procedures such as patching, automated backups, scaling, monitoring, etc.

Refactor / Re-architect

Refactoring is often the most expensive strategy for customers and is also the most complex. Although this strategy allows you to reap the most out of AWS, there are a lot of factors and decision-making involved which can prolong your migration process. Sometimes, it is easier to just start fresh rather than modifying legacy systems to fit into AWS. Another option that you can do is to first rehost or replatform as much of your system as you can, and slowly but surely refactor them to fit your desired environment.

It takes a lot of expertise and understanding of different technologies, not just AWS, to properly architect a system in the cloud. This is why Solutions Architects are paid handsomely in the industry. AWS is continuously growing and innovating new products, and as a Solutions Architect, you need to be up-to-date with these



offerings to bring the best value to your customers. Hence, certifications are a must if you want to reach the highest level of this craft.

Repurchase

This strategy discusses moving from perpetual licenses to a software-as-a-service model. Figure out what products are available out there that you can adopt instead of using your own systems. This reduces the chances of incompatibility and allows you to focus on your value-adding operations. By purchasing or adopting well-known alternatives, you also gain access to a larger user base with better support and more consistent updates from the developers of the product.

Retain

Although this strategy can be a bit counterintuitive, performing a migration doesn't necessarily mean you have to move *everything immediately*. Sometimes, it can be better to leave an application behind or to hold off its migration temporarily. For example, if you only allocated a specific budget for this task, you can start off with migrating crucial applications first, and leave the rest on-prem. This way, you can strictly control the whole migration procedure and design better implementations as you go. The only downside to this strategy is that it can prolong your whole migration journey, which in effect can increase the total cost and effort than what is necessary.

Solutions Architect Professional Exam Notes:

It can sometimes be impossible to move a system to AWS without fully re-architecting it. An application can be very old and too deeply rooted in one's current operations to properly move it to the cloud. Other customers may cite that they need their applications to stay on-premises due to compliance reasons, or because their (expensive) software licenses are tied to their servers. There are also a lot of customers who go for a hybrid environment because they do not want to fully commit to the cloud. Be sure to take note of your scenarios to know when to retain and to not retain applications.

Strategy (increasing complexing)	Effort and Cost	Opportunity to Optimize
Retire	N/A	N/A
Retain	Minimal effort and cost	N/A
Relocate	Minimal effort and cost	Small
Rehost	Minimal to Average effort and cost	Small
Repurchase	Average effort and cost	Small



Replatform	Above average effort and cost	Average
Refactor/Re-architect	High effort and cost	High

References:

<https://docs.aws.amazon.com/prescriptive-guidance/latest/migration-retiring-applications/overview.html>
<https://docs.aws.amazon.com/prescriptive-guidance/latest/application-portfolio-assessment-guide/iterating-7-rs-migration-strategy-selection.html>

Analyzing Your Workloads Using AWS Application Discovery Service

When you have hundreds to thousands of servers, virtual machines, and applications running in your on-premises infrastructure, it can be tedious to do an inventory and analyze all their usage patterns. Data collection and dependency mapping are very important tasks during the migration planning phase, as it will influence your decision-making and the outcome of your migration process. To conduct this analysis in a simpler manner, you can use AWS Application Discovery Service to perform these tasks for you.

AWS Application Discovery Service is an automated solution that collects and presents configuration, usage, and behavior data from your servers to help you better understand your workloads. These data are then stored in the AWS Application Discovery Service local data store, and can be exported in csv format. The data will help you estimate the Total Cost of Ownership (TCO) of running on AWS. When paired with AWS Migration Hub, you can use the resulting data to migrate the discovered servers and applications using an AWS or partner migration tool, and track their progress as they get migrated to AWS.

AWS Application Discovery Service works in two ways depending on the environment to be scanned. If you are a VMware user, AWS Application Discovery Service uses an agentless discovery process to collect VM configuration and performance profiles. Users in a non-VMware environment or those that need additional information, such as network dependencies and information about running processes, will need to install the Application Discovery Agent on each of your servers and VMs.

AWS Application Discovery Service is able to collect the following information:

- Server hostnames,
- IP addresses,
- MAC addresses,
- CPU, network, memory, and disk utilization
- Disk and network performance (e.g., latency and throughput)

AWS Application Discovery Service agents record inbound and outbound network activity for each server. This data can then be used to understand the dependencies across servers. For VMware environments, you won't be able to record if you do not install the agent first.



Do note that the AWS Application Discovery Service does not perform any type of migration. It is purely a discovery service that integrates well with other AWS migration services.

Reference:

<https://aws.amazon.com/application-discovery>

<https://aws.amazon.com/migration-hub/>

[How AWS Migration Hub Helps You Plan, Track, and Complete Your Application Migrations](#)

Performing Data Migration

i Solutions Architect Professional Exam Notes:

How much data can move from your on-premises data center to AWS through your current network connection? For a best case scenario, you can use this formula:

$$\begin{aligned} \text{No. of days} &= (\text{Total Bytes}) / \left(\begin{aligned} &(\text{Megabits per second} * 125 * 1000) \\ &* \text{Network Utilization} \\ &* (60 \text{ seconds} * 60 \text{ minutes} * 24 \text{ hours}) \end{aligned} \right) \end{aligned}$$

For example:

- You have a T1 connection (1.544Mbps) with a network utilization of 80% and 1 TB of data to move in or out of AWS. 1 TB is equivalent to 1,099,511,627,776 bytes. Using the formula above, we'll get the following result:

$$\begin{aligned} &= (\text{Total Bytes}) / (\text{Megabits per second} * 125 * 1000) * \text{Network Utilization} * \text{Time} \\ &= (1,099,511,627,776) / (1.544 * 125 * 1000) * (0.80) * (60 * 60 * 24) \\ &= 1,099,511,627,776 / 13,340,160,000 \\ &= 82.42 \end{aligned}$$

- As calculated above, the theoretical minimum time it would take to load over your network connection at 80% network utilization is **82** days. Sometimes, you might not have a calculator on-hand, so you can always go for rough estimates instead.

AWS Storage Gateway lets you connect and extend your on-premises applications to AWS storage services such as S3, S3 Glacier and EBS.

- File Gateway uses **SMB or NFS file shares** for on-premises applications to store files as **S3 objects** and access them with traditional file interfaces.



- Volume Gateway stores or caches block volumes locally, with point-in-time backups as **EBS snapshots**. These snapshots can also be recovered in the cloud.
- Tape Gateway **virtual tape library** (VTL) configuration integrates with existing backup software for cost effective tape replacement in **Amazon S3** and long term archival in **S3 Glacier Flexible Retrieval** and **S3 Glacier Deep Archive**.

Use AWS Storage Gateway if you need to sync appliances with Amazon S3 or generate Amazon EBS volumes. Transfer speeds will depend on your network connection speed.

AWS Direct Connect is a dedicated physical connection to accelerate network transfers between your datacenters and AWS datacenters. This dedicated connection can be partitioned into multiple virtual interfaces. Partitioning enables you to use the same connection to access public resources such as objects stored in Amazon S3 using public IP address space, and private resources such as Amazon EC2 instances running within a VPC using private IP space, while maintaining network separation between the public and private environments. **1Gbps and 10Gbps ports** are available. You can order lines with transfer speeds of **50Mbps, 100Mbps, 200Mbps, 300Mbps, 400Mbps, and 500Mbps**. Direct Connect works with both **IPsec VPN** using public VIF and **AWS Transit Gateway** with transit VIF if you need to connect to multiple VPCs.

AWS S3 Transfer Acceleration makes public Internet transfers to Amazon S3 faster by taking advantage of **Amazon CloudFront's globally distributed edge locations**. There is **no guarantee** that you'll experience better transfer speeds, so if you need consistent, fast transfers, use other options. If you need a low cost option in speeding up transfers whenever acceleration is available, this will do.

AWS Data Sync is a data transfer service that **automates** moving data between on-premises storage and **Amazon S3, Amazon EFS, or Amazon FSx for Windows File Server**. You can use DataSync to copy data over AWS Direct Connect or Internet links to AWS for data migrations, recurring data processing workflows, and automated replication for data protection and recovery.

Amazon Data Firehose is the easiest way to capture, modify, and deliver data streams into various AWS services such as **Amazon S3, Amazon Redshift, Amazon OpenSearch Service, Splunk, Snowflake**, and other 3rd party analytics services. Additionally, it simplifies the process of maintaining streaming data delivery pipelines by managing the provisioning and scaling of resources for you.

References:

<https://aws.amazon.com/cloud-data-migration/>

<https://docs.aws.amazon.com/whitepapers/latest/aws-vpc-connectivity-options/aws-direct-connect-vpn.html>

[Migrating Data to AWS: Understanding Your Options - AWS Online Tech Talks](#)



Performing Database Migration

For database migration, you will be using AWS Database Migration Service if you need a managed solution without giving your database downtime. **AWS Database Migration Service** (AWS DMS) can migrate your data to and from relational databases, data warehouses, NoSQL databases, and other types of data stores. You can perform homogeneous migrations such as Oracle to Oracle, and heterogeneous migrations between different database platforms, such as Oracle to Amazon Aurora or Microsoft SQL Server to MySQL. It also allows you to stream data to Amazon Redshift from any of the supported sources including Amazon Aurora, PostgreSQL, MySQL, MariaDB, Oracle, SAP ASE, and SQL Server.

AWS Database Migration Service can also be used for continuous data replication with high availability through change data capture (CDC) to keep your databases in-sync even after migration has completed.

To perform a database migration, AWS DMS connects to the source data store, reads the source data, and formats the data for consumption by the target data store. It then loads the data into the target data store. Cached transactions and log files are also written to disk.

At a high level, you perform the following to initiate a migration:

- Create a replication server
- Create source and target endpoints that have connection information about your data stores
- Create one or more migration tasks to migrate data between the source and target data stores

Once migration has started:

- AWS DMS will first do a full migration where existing data from the source is moved to the target. While this is in progress, any changes made to the tables being loaded are cached on the replication server.
- Once the full migration is finished, AWS DMS will apply the cached changes stored in the replication server.
- When all tables have been loaded on the target, AWS DMS begins to collect changes as transactions for the ongoing replication phase.

If your migration is heterogeneous (between two databases that use different engine types), you can use the **AWS Schema Conversion Tool** (AWS SCT) to generate a complete target schema for you. If you use the tool, any dependencies between tables, such as foreign key constraints, need to be disabled during the migration's "full load" and "cached change apply" phases.

Oftentimes, we associate migration with moving things into AWS, but this is not always the case. There are cases when you are asked to migrate or sync a database outside of AWS, and with the least possible downtime. If you are hosting your database in Amazon RDS for MySQL and you also have an on-premises MySQL server, you can migrate data from Amazon RDS for MySQL to your on-premises database server.



To do so:

- Create an Amazon RDS for MySQL read replica
- Switch the replication target from the read replica to the on-premises server
- Once replication is finished and the pair are in-sync, you may delete the read replica

If none of these options work for you, there is always the traditional backup and restore route. However, it will be difficult to keep your databases in-sync as new changes come in and experience little downtime during the switchover.

References:

<https://docs.aws.amazon.com/whitepapers/latest/aws-overview/migration-services.html>

<https://docs.aws.amazon.com/dms/latest/userguide>

<https://aws.amazon.com/premiumsupport/knowledge-center/replicate-amazon-rds-mysql-on-premises/>